If $a = 7,\ d = -2$ then the four numbers are 13, 9, 5 and 1

Therefore, the four consecutive terms of the A.P. are 1, 5, 9 and 13.

Condition for three numbers to be in A.P.

If a, b, c are in A.P. then $a = a,\ b = a + d,\ c = a + 2d$

$$\text{so} \quad a + c = 2a + 2d = 2(a + d) = 2b$$

$$\text{Thus, } 2b = a + c$$

Similarly, if $2b = a + c$, then $b - a = c - b$ so a, b, c are in A.P.

Thus three non-zero numbers a, b, c are in A.P. if and only if $2b = a + c$

Example 2.30 A mother divides ₹207 into three parts such that the amount are in A.P. and gives it to her three children. The product of the two least amounts that the children had ₹4623. Find the amount received by each child.

Solution Let the amount received by the three children be in the form of A.P. is given by $a - d,\ a,\ a + d$. Since, sum of the amount is ₹207, we have

$$(a - d) + a + (a + d) = 207$$

$$3a = 207 \Rightarrow a = 69$$

It is given that product of the two least amounts is 4623.

$$(a - d)a = 4623$$

$$(69 - d)69 = 4623$$

$$d = 2$$

Therefore, amount given by the mother to her three children are

₹(69−2), ₹69, ₹(69+2). That is, ₹67, ₹69 and ₹71.

Progress Check

1. If every term of an A.P. is multiplied by 3, then the common difference of the new A.P. is _____.
2. Three numbers a, b and c will be in A.P. if and only if _____.

Exercise 2.5

1. Check whether the following sequences are in A.P.

   (i) $a - 3, a - 5, a - 7, \ldots$ (ii) $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$ (iii) 9, 13, 17, 21, 25,...

   (iv) $\frac{-1}{3}, 0, \frac{1}{3}, \frac{2}{3}, \ldots$ (v) $1, -1, 1, -1, 1, -1, \ldots$

2. First term a and common difference d are given below. Find the corresponding A.P.

   (i) $a = 5,\ d = 6$ (ii) $a = 7,\ d = -5$ (iii) $a = \frac{3}{4},\ d = \frac{1}{2}$

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